

Proof by induction - Matrices

Basic skills

Q1. $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

a) $A^2 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$

dot Product row vec with col vec

b) $A^3 = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}$

c) $A^4 = \begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 16 & 0 \\ 0 & 1 \end{pmatrix}$

Q2. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

a) $A^2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$

b) $A^3 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$

c) $A^4 = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$

Q3. Has 'nth term' so no need to calculate prior powers

a) $A^2 = \begin{pmatrix} 2 \times 2 + 1 & 3 \times 2 \\ 2 + 5 & 2 \times 2 - 1 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 7 & 3 \end{pmatrix}$

$$b) A^3 = \begin{pmatrix} 2 \times 3 + 1 & 3 \times 3 \\ 3 + 5 & 2 \times 3 - 1 \end{pmatrix} = \begin{pmatrix} 7 & 9 \\ 8 & 5 \end{pmatrix}$$

$$c) A^{10} = \begin{pmatrix} 2 \times 10 + 1 & 3 \times 10 \\ 10 + 5 & 2 \times 10 - 1 \end{pmatrix} = \begin{pmatrix} 21 & 30 \\ 15 & 19 \end{pmatrix}$$

Q4. $\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 4n+2 & n-1 \\ 5 & 2n \end{pmatrix}$

$$= \begin{pmatrix} 2 \times (4n+2) + 1 \times 5 & 2(n-1) + 2n \\ 1 \times (4n+2) & n-1 \end{pmatrix}$$

$$= \begin{pmatrix} 8n+9 & 4n-2 \\ 4n+2 & n-1 \end{pmatrix}$$

Competency skills

Q 5. $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$

a) Base case: sub n=1

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ so yes}$$

b)

$$A^{k+1} = A^k A$$

↓ Assumption

$$= \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$$

c) As $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ is true for $n=1$

and if $A^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ then $A^{k+1} = \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$

we have $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for all n , by induction

Q 6. $A = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ $A^n = \begin{pmatrix} 2^n & 0 \\ 0 & \frac{1}{2^n} \end{pmatrix}$

Base case: $\begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 2^1 & 0 \\ 0 & \frac{1}{2^1} \end{pmatrix}$ so holds

I.H: Assume $A^k = \begin{pmatrix} 2^k & 0 \\ 0 & \frac{1}{2^k} \end{pmatrix}$ for some k

I.S:

$$A^{k+1} = A^k A$$

↓ I.H

$$= \begin{pmatrix} 2^k & 0 \\ 0 & \frac{1}{2^k} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 2 \times 2^k & 0 \\ 0 & \frac{1}{2^k} \times \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 2^{k+1} & 0 \\ 0 & \frac{1}{2^{k+1}} \end{pmatrix}$$

Conclusion:

Formula true for $n=1$ and if true for k then true for $k+1$, hence by induction it is true for any n .

Q7. $A = \begin{pmatrix} 3 & 1 \\ -4 & -1 \end{pmatrix}$ $A^n = \begin{pmatrix} 1+2n & n \\ -4n & 1-2n \end{pmatrix}$

Base case:

$$\begin{pmatrix} 3 & 1 \\ -4 & -1 \end{pmatrix} = \begin{pmatrix} 1+2 & 1 \\ -4 \times 1 & 1-2 \times 1 \end{pmatrix}$$

So holds

I.H: Assume formula holds for some k

I.S:

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 1+2k & k \\ -4k & 1-2k \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -4 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 3(1+2k) - 4k & 1+2k - k \\ -12k - 4(1-2k) & -4k - (1-2k) \end{pmatrix}$$

$$= \begin{pmatrix} 2k+3 & 1+k \\ -4k-4 & -2k-1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+2(k+1) & k+1 \\ -4(k+1) & 1-2(k+1) \end{pmatrix}$$

Conclusion:

Formula true for $n=1$ and if true for k then true for $k+1$, hence by induction it is true for any n .

Advanced skills

Q8. $A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ then

$$A^2 = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2a \\ 0 & 1 \end{pmatrix} \quad \text{Predict}$$

$$A^3 = \begin{pmatrix} 1 & 2a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3a \\ 0 & 1 \end{pmatrix} \quad A^n = \begin{pmatrix} 1 & na \\ 0 & 1 \end{pmatrix}$$

Base case: $\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \times a \\ 0 & 1 \end{pmatrix}$

I. H: Assume $A^k = \begin{pmatrix} 1 & k a \\ 0 & 1 \end{pmatrix}$ for some k

I. S: $A^{k+1} = A^k A = \begin{pmatrix} 1 & k a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 1 & a + k a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & (k+1)a \\ 0 & 1 \end{pmatrix}$$

Conclusion:

Formula true for $n=1$ and if true for k then true for $k+1$, hence by induction it is true for any n .

Q9. $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ $A^n = \begin{pmatrix} 1 & n & \frac{1}{2}n(n-1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

Base case: $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & \frac{1}{2} \times 1 \times 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ holds

I.H: Assume $A^k = \begin{pmatrix} 1 & k & \frac{1}{2}k(k-1) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$

I.S:

$$A^{k+1} = A^k A$$

$$= \begin{pmatrix} 1 & k & \frac{1}{2}k(k-1) \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1+k & k+\frac{1}{2}k(k-1) \\ 0 & 1 & 1+k \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k+1 & \frac{1}{2}(k+1)(k) \\ 0 & 1 & k+1 \\ 0 & 0 & 1 \end{pmatrix}$$

Conclusion:

Formula true for $n=1$ and if true for k then true for $k+1$, hence by induction it is true for any n .

Q10.

$$A = \begin{pmatrix} 4 & -3 & -3 \\ 3 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}$$

a) To diagonalise need eigenvalues and eigenvectors

$$\begin{vmatrix} 4-\lambda & -3 & -3 \\ 3 & -2-\lambda & -3 \\ -1 & 1 & 2-\lambda \end{vmatrix} = (4-\lambda) \begin{vmatrix} -2-\lambda & -3 \\ 1 & 2-\lambda \end{vmatrix}$$

$$- -3 \begin{vmatrix} 3 & -3 \\ -1 & 2-\lambda \end{vmatrix} + -3 \begin{vmatrix} 3 & -2-\lambda \\ -1 & 1 \end{vmatrix}$$

$$= (4-\lambda) ((-2-\lambda)(2-\lambda) + 3) \\ + 3 (3(2-\lambda) - 3) - 3 (3 - (2+\lambda))$$

$$= -(4-\lambda)(2-\lambda)(2+\lambda) + 12 - 3\lambda + 18 - 9\lambda - 9 - 9 + 6 + 3\lambda$$

$$= -((\lambda^2 - 6\lambda + 8)(2 + \lambda)) + 18 - 9\lambda$$

$$= -(\lambda^3 - 4\lambda^2 - 4\lambda + 16) + 18 - 9\lambda$$

$$= -\lambda^3 + 4\lambda^2 - 5\lambda + 2$$

For eigenvalues

$$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

$\lambda = 2$ is a solution as

$$8 - 4 \times 4 + 5 \times 2 - 2 = 0$$

Factorise

$$(\lambda - 1)^2(\lambda - 2) = 0$$

eigenvalues are $\lambda = 1, 1, 2$

For eigenvectors solve $\lambda v = Av$

For $\lambda = 1$

$$1 \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 & -3 & -3 \\ 3 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\text{so } x = 4x - 3y - 3z$$

$$y = 3x - 2y - 3z$$

$$z = -x + y + 2z$$

solving gives

$$x = y + z$$

notice 2 free variables

so two vectors

$$\text{e.g. } \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

For $\lambda = 2$

$$2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 & -3 & -3 \\ 3 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$2x = 4x - 3y - 3z$$

$$2y = 3x - 2y - 3z$$

$$2z = -x + y + 2z$$

solving gives

$$x = -3z$$

$$y = -3z$$

so vector $\begin{pmatrix} -3 \\ -3 \\ 1 \end{pmatrix}$
e.g.

$$\lambda = 1 \quad \lambda = 1 \quad \lambda = 2$$

so

$$S = \begin{pmatrix} 1 & 1 & -3 \\ 1 & 0 & -3 \\ 0 & 1 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

can check $A = SDS^{-1}$

b) $D^n = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^n \end{pmatrix}$ ← to power a diag can just power diag.

can prove by induction

This can be used, here we prove it

base case is true as

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^1 \end{pmatrix} = D$$

I.H: Assume works for some k

I.S:

$$D^{k+1} = D^k D$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^k \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 1^2 & 0 & 0 \\ 0 & 1^2 & 0 \\ 0 & 0 & 2 \times 2^k \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^{k+1} \end{pmatrix}$$

So works by induction and is true for all n .

c)

$$A^n = (S D S^{-1})^n$$

$$= S D \cancel{S^{-1} S} D \cancel{S^{-1} S} \dots \cancel{S D S^{-1}}$$

$$= S D D \dots D S^{-1} = S D^n S^{-1}$$

need S^{-1} to compute.

$$S = \begin{pmatrix} 1 & 1 & -3 \\ 1 & 0 & -3 \\ 0 & 1 & 1 \end{pmatrix} \text{ by cofactor method}$$

$$S^{-1} = \frac{1}{|S|} \begin{pmatrix} +|M_{11}| & -|M_{12}| & +|M_{13}| \\ -|M_{21}| & +|M_{22}| & -|M_{23}| \\ +|M_{31}| & -|M_{32}| & +|M_{33}| \end{pmatrix}$$

$$= \frac{1}{-1} \begin{pmatrix} 3 & -4 & -3 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 4 & 3 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$$

$$\text{So } A^n = \begin{pmatrix} 1 & 1 & -3 \\ 1 & 0 & -3 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2^n \end{pmatrix} \begin{pmatrix} -3 & 4 & 3 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 & -3 \times 2^n \\ 1 & 0 & -3 \times 2^n \\ 0 & 1 & 2^n \end{pmatrix} \begin{pmatrix} -3 & 4 & 3 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 + 1 + 3 \times 2^n & 4 - 1 - 3 \times 2^n & 3 - 3 \times 2^n \\ -3 + 3 \times 2^n & 4 - 3 \times 2^n & 3 - 3 \times 2^n \\ 1 - 2^n & -1 + 2^n & 2^n \end{pmatrix}$$

$$= \begin{pmatrix} -2 + 3 \times 2^n & 3 - 3 \times 2^n & 3 - 3 \times 2^n \\ -3 + 3 \times 2^n & 4 - 3 \times 2^n & 3 - 3 \times 2^n \\ 1 - 2^n & 2^n - 1 & 2^n \end{pmatrix}$$